ANTO VARGHESE



Software Engineer (C++ / Go / AI Workflows)

Anto Varghese

Building and scaling high-performance, multithreaded systems in C, C++, and Go, accelerated with modern AI tooling & prompt engineering. Experienced in monolith-to-microservices modernization, retail Point-of-Sale engines, and real-time traffic monitoring systems.

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PROFESSIONAL SUMMARY

7+ years of engineering robust, event-driven backends.

Specialized in low-level systems programming, memory-sensitive execution, and high-concurrency environments. Demonstrated history in maintaining and decoupling large

monolithic codebases into scalable, modern microservices.

Extensive experience collaborating across cross-functional teams, producing flow specifications, and interfacing software with critical real-world infrastructure like ANPR hardware, speed enforcement cameras, and retail Point-of-Sale terminals.

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LOCATION: Konni, Kerala, India
CURRENT: Software Engineer @ UST
PREVIOUS: ARS Traffic & Transport Tech
DEGREE: B.E. in Electronics & Communication
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TECHNICAL ARSENAL

Core Competencies

C / C++

Deep experience with STL, multithreading, Windows Services, MFC, and low-level system design.

Go (Golang)

Developing concurrent, high-throughput backend services and modern microservice APIs.



AI & Prompt Engineering

Prompt engineering, LLM workflow automation, API integration, and AI-assisted code generation & debugging in VS Code.



Microservices & Migration

Decomposing complex monolithic legacy event-based systems into modular, independently deployable services.



Databases & Storage

Relational modeling, query optimization, and transaction management using MySQL and MS SQL Server.



Embedded & Hardware Interfacing

Embedded C/C++, Arduino robotics, sensor integration, and real-time passage detection logic.

CAREER TRAJECTORY

Work Experience

7+ YEARS TOTAL

Software Engineer @ UST

Jun 2024 - Present | Thiruvananthapuram

Project: Keele (Retail Point-of-Sale)

- Architected and optimized high-throughput Point of Sale backend components in C and Go.
- Stabilized mission-critical event-based monolithic systems, mitigating production crashes and latency spikes.
- Led the decomposition of monolithic legacy architecture into maintainable, scalable microservices.
- Constructed end-to-end data flow specifications and aligned development objectives with client requirements.

C | GO | MICROSERVICES | AI PROMPTING | POS SYSTEMS

Software Engineer @ ARS Traffic & Transport Tech

Feb 2020 - May 2024 | Technopark Trivandrum

Projects: VACAm & TracoOWN, WIM, NDW, TVM SmartCity

- Designed multithreaded Windows Services in VC++ utilizing MFC, STL, and SQL engines for automated passage capture.
- Implemented calculation engines for the VACAm Average Speed Enforcement System to detect trajectory violations between check gates.
- Engineered Weigh-In-Motion (WIM) modules measuring real-time axle weight distribution to preserve road structural integrity.
- Built high-reliability data capture services for the National Data Warehouse (NDW) traffic forecasting system.

C++ | MULTITHREADING | WINDOWS SERVICES | MFC / STL | SQL

KEY HIGHLIGHTS

Featured Projects

EMBEDDED AUTOMATION

Voice Controlled Automation

Integrated microcontroller automation enabling seamless voice recognition control over household and electronic utilities.

VIEW PROJECT SPECIFICATIONS →

SMART LOGGING

Automatic Attendance System

Arduino-based identification logging system utilizing RFID reader sensors for automated student check-ins.

VIEW PROJECT SPECIFICATIONS →

ANTO VARGHESE

Specializing in high-performance backends, low-level systems, and AI workflows.

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